12TH CHEMISTRY · CHAPTER 33 · SOLVED EXERCISE

Water

Complete textbook exercise (MCQs, Short-Answer, Constructed Response, Descriptive) with Quick Check solutions

Q1. MULTIPLE CHOICE QUESTIONS

Four choices are given for each questions. Select the correct choice.

I. Which property of water is primarily responsible for its ability to resist rapid temperature changes in large aquatic ecosystems?

(a) Amphoteric nature
(b) High specific heat capacity
(c) The permanent dipole
(d) High redox potential

Answer: (b) High specific heat capacity — Hydrogen bonding gives water a high specific heat capacity, so large water bodies are thermally stable.

II. A water sample is found to have a very high Biochemical Oxygen Demand (BOD). What does this most likely indicate about the water body?

(a) Highly oxygenated and healthy
(b) High concentration of nitrates and phosphates.
(c) High organic pollution
(d) A high pH

Answer: (c) High organic pollution — High BOD means much organic matter to be broken down by microorganisms.

III. Which of the following best describes the formation of a "pollution plume"?

(a) Agricultural runoff
(b) Discharge of heavy metal salts
(c) The mixture of hydrocarbons and road salts
(d) Snowmelt through uncontaminated forest soil.

Answer: (b) Discharge of heavy metal salts — A plume forms where a point source (e.g. a factory pipe releasing lead or mercury salts) discharges at one place.

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IV. Why is Non-Point Source (NPS) pollution considered "weather-dependent" compared to Point Source pollution?

(a) NPS pollutants are only toxic when the temperature of the water rises.
(b) NPS pollution requires precipitation to wash diffuse surface contaminants into water bodies.
(c) Point sources only discharge waste during specific industrial seasons.
(d) Rainfall neutralizes the localized acute toxicity of point source plumes.

Answer: (b) NPS pollution requires precipitation to wash diffuse surface contaminants into water bodies. — Rainfall or snowmelt carries the scattered pollutants into the water.

V. Bacteria in aquatic sediments convert inorganic mercury into methylmercury ($[CH_3Hg]^+$). Which statement best explains why this conversion increases the ecological hazard?

(a) Methylmercury dissolves more readily in the water
(b) Methyl mercury is biodegradable and enters nutrient cycle.
(c) Methylmercury is highly fat-soluble
(d) Methylmercury is a solid that clogs the gills of fish.

Answer: (c) Methylmercury is highly fat-soluble — It crosses biological membranes and is biomagnified up the food chain.

VI. Which condition is a direct result of Cadmium (Cd) replacing Calcium (Ca) in the skeletal system?

(a) Minamata disease
(b) Blue Baby Syndrome
(c) Itai-itai disease
(d) Chloracne

Answer: (c) Itai-itai disease — Cadmium replaces calcium in bones, making them extremely brittle.

VII. What is the primary chemical role of Aluminium Sulphate ($Al_2(SO_4)_3$) in the wastewater treatment process?

(a) To kill pathogenic bacteria.
(b) To act as a semi-permeable membrane.
(c) Converting colloidal particles to floc.
(d) To adsorb organic contaminants.

Answer: (c) Converting colloidal particles to floc. — It is a coagulant: small particles clump into larger masses (floc).

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VIII. Ion exchange technology "softens" water by replacing calcium (Ca^{2+}) and magnesium (Mg^{2+}) ions with:

(a) Chlorine (Cl_2) or ozone (O_3)
(b) Sodium (Na^+) or hydrogen (H^+) ions
(c) Aluminum (Al^{3+}) or iron (Fe^{3+}) ions
(d) Carbonate (CO_3^{2-}) or phosphate (PO_4^{3-}) ions

Answer: (b) Sodium (Na^+) or hydrogen (H^+) ions — In the ion-exchange column the hardness ions are swapped for Na^+ or H^+.

Q2. SHORT-ANSWER QUESTIONS

a) Differentiate between Point Source and Non-Point Source pollution in terms of the "concentration" of pollutants at the site of entry.

In point-source pollution the discharge is concentrated at one point, so the pollutant concentration is highest at the site of entry or just downstream (a pollution plume with localised acute toxicity). In non-point source pollution a mixture of pollutants enters at lower concentrations spread over a much wider area.

b) Using an example from the text, explain how an underground storage tank can become a point source of pollution.

If a storage tank at a chemical plant leaks, its contents pass into the soil and then into the groundwater. The pollution enters from one single, identifiable location (the tank), so it is a point source.

c) Identify two types of chemical pollutants commonly found in urban storm water runoff.

Leaked engine oils (hydrocarbons) and zinc from tyres (a heavy metal); road salts are also present.

d) Differentiate between the "solubility" of pesticides like DDT and inorganic nutrients like nitrates in the context of the water cycle.

Nitrates are ionic, so they dissolve readily in polar water and are carried in solution through the water cycle (causing eutrophication). DDT is non-polar and insoluble in water; it is carried by run-off and dissolves in the fatty tissues of fish and birds, where it accumulates.

e) Explain why microplastics are considered a "growing environmental crisis" differently than organic solid waste.

Organic solid waste biodegrades. Plastics do not biodegrade; they only break down into smaller particles (microplastics) that persist in the environment for centuries. Microplastics are eaten by marine animals and have entered the food chain, raising concerns for human health.

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f) How an oil spill physically disrupts the chemical balance of a marine ecosystem?

Oil spreads rapidly over the water surface and forms a layer. This layer blocks sunlight (so photosynthesis decreases) and reduces the exchange of oxygen between air and water, so the dissolved oxygen falls and marine life is harmed.

g) Why are synthetic organic pollutants, like PCBs and pesticides, considered "persistent" in the environment?

They are not easily broken down (non-biodegradable), so they remain in the environment for a long time. Being non-polar, they are insoluble in water but dissolve in fatty tissues, so they are not washed out of organisms and accumulate in the food chain.

h) How do the decomposition of organic matter in polluted water contributes to global warming?

Sewage and run-off boost the growth of algae and other organisms. When they die and decompose in oxygen-poor (anaerobic) conditions, methane is released. Methane is a potent greenhouse gas that worsens global warming.

i) Name four methods used for water conservation.

(1) Drip and sprinkler irrigation. (2) Reusing and recycling greywater. (3) Rainwater harvesting. (4) Regular repair of the water-supply system to stop leaks. (Also: small dams and recharge wells; public awareness.)

j) What is the role of "industrial discharge" in water pollution?

Industrial activities are the major source of water pollution. Factories such as oil refineries, chemical plants, tanneries and fertilizer plants discharge harmful chemicals, heavy metals (e.g. lead and mercury salts) and toxic waste directly into rivers, lakes and oceans, usually as point sources.

k) Both reverse osmosis (RO) and ion exchange methods are used to obtain high-purity water?
a) Differentiate between the processes involved in both these methods.
b) What impurities do these methods target?

a) In reverse osmosis, water is forced under pressure through a semi-permeable membrane, which filters out the impurities. In ion exchange, water passes through a column in which unwanted ions are swapped for other ions.

b) RO removes dissolved salts, bacteria and other impurities. Ion exchange removes the hardness ions Ca^{2+} and Mg^{2+}, replacing them with Na^{+} or H^{+} ions.

Q3. CONSTRUCTED RESPONSE QUESTIONS

a) Explain the relationship between acid rain and the presence of toxic heavy metals in aquatic environments.

Acid rain is formed when SO_2 and nitrogen oxides dissolve in rain water; it increases the concentration of hydronium ions (H_3O^+) in water bodies. This acidity mobilises toxic heavy-metal compounds, such as aluminium hydroxide and lead salts, from the sediments into the water: their solubility increases through protonation, or the metal ions are displaced from mineral surfaces. The dissolved metal ions are then available to aquatic organisms.

$$Al(OH)_{3(s)} + 3H_3O^+_{(aq)} \rightarrow Al^{3+}_{(aq)} + 6H_2O_{(l)}$$

The equation illustrates the protonation described in the text; it is not printed in the chapter.

b) Analyze the environmental impact of agricultural runoff as a Non-Point Source of pollution. Reference specific chemicals in your answer.

Rain washes chemicals from thousands of hectares of farmland into river systems; the pollution is diffuse, weather-dependent and spread over a wide area.

- **Fertilizers (ammonium nitrate NH_4NO_3, phosphates):** nitrogen and phosphorus cause eutrophication: algal blooms, oxygen depletion and dead zones; nitrates in drinking water cause Blue Baby Syndrome.
- **Pesticides and herbicides (DDT, glyphosate):** persistent organic pollutants that dissolve in fatty tissue, bioaccumulate, act as endocrine disruptors and cause eggshell thinning in birds.
- **Cadmium** from phosphate fertilizer impurities accumulates in the food chain.

c) Evaluate the role of agricultural runoff" in contributing to different types of water pollution.

Agricultural run-off contributes to several types of water pollution at once:

Type of pollution	Contribution of agricultural run-off
Nutrient pollution (eutrophication)	Fertilizers supply nitrogen and phosphorus.
Chemical pollution	Pesticides, herbicides and cadmium from phosphate fertilizers.
Microbiological pollution	Animal waste carries bacteria, viruses and protozoa.
Sediment pollution	Poor irrigation increases soil erosion, washing sediments into rivers.

It is therefore a major, significant contributor; because it is a non-point source spread over wide areas, it is also difficult to control. It contaminates both surface water and groundwater.

d) Evaluate the benefits and the long-term chemical risks involved in the use of chlorine as a disinfectant.

Benefits: chlorination is the most common method of disinfection. Chlorine forms hypochlorous acid (HOCl), which penetrates the cell walls of bacteria and viruses and disrupts their functions, so water-borne diseases such as cholera and typhoid are prevented.

Long-term risks: chlorine reacts with organic matter in water to form chlorinated organic compounds (e.g. trihalomethanes such as chloroform), which are harmful on long-term exposure and are suspected carcinogens. Chlorine gas itself is toxic and must be handled carefully. The dose must therefore be controlled.

The chapter describes only the disinfecting action of chlorine; the long-term risks are from general knowledge.

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DESCRIPTIVE QUESTIONS

Q4. Describe the effects of heavy metals and pesticides on the human body.

Heavy metals

Heavy metals have high atomic weights and densities. They are non-biodegradable and accumulate in the tissues of living organisms.

Metal	Effect on the human body
Mercury (as methylmercury)	Highly fat-soluble, so it crosses biological membranes and the blood–brain barrier and is biomagnified. A potent neurotoxin; long exposure causes **Minamata disease** (ataxia, numbness in hands and feet, muscle weakness).
Lead	Interferes with the production of haemoglobin, causing anaemia. In children it is a cumulative poison causing irreversible neurological damage and developmental delays.
Cadmium	Long-term ingestion causes **Itai-itai disease**: Cd^{2+} replaces Ca^{2+} in bones, making them extremely brittle. It also causes severe kidney damage (nephrotoxicity).

Pesticides (and PCBs)

- Many pesticides act as **endocrine disruptors**, behaving like natural hormones such as estrogen; they are linked to hormonal disruption, developmental problems and hormonal cancers in humans.
- In wildlife they cause reproductive failure, e.g. thinning of eggshells in birds.
- PCBs are probable human carcinogens; they also suppress the immune system and cause skin conditions such as chloracne.
- Chemical pollution also degrades water quality, reduces biodiversity and disrupts aquatic ecosystems.

Q5. Explain the different sources of heavy metals and organic compounds.

Heavy metals

Toxic heavy metals such as mercury, lead, arsenic and cadmium enter water through industrial discharge, mining operations and improper waste disposal.

Metal	Sources
Mercury	Coal combustion, gold mining, industrial waste from thermometer or battery manufacturing. In sediments, bacteria convert inorganic mercury into the more dangerous methylmercury, $[CH_3Hg]^+$.
Lead	Leaded petrol (now banned worldwide), old plumbing (lead pipes), lead-acid batteries and paints. Pb^{2+} ions often enter water through corrosion of pipes.
Cadmium	Metal plating, plastic stabilisers and impurities in phosphate fertilizers.

These pollutants are non-biodegradable and accumulate in the food chain, posing long-term risks to aquatic life and humans.

Organic compounds

Pesticides and herbicides such as DDT and glyphosate come mainly from agricultural run-off and domestic gardening. They are non-polar and insoluble in water but highly soluble in the fatty tissues of fish and birds. Many older pesticides like DDT are **persistent organic pollutants (POPs)**. Polychlorinated biphenyls (PCBs), used as coolants in electrical transformers, leak into water from hazardous waste sites.

Other organic pollutants: oil (hydrocarbons) from ship leaks and spills and from engine oil in road run-off; organic waste from sewage.

Q6. Describe various technologies used in water treatment.

Different methods are used according to the type and severity of pollution (organic matter, chemicals or pathogens). They are applied at three levels: **primary treatment** (removal of physical debris), **secondary treatment** (biological processing) and **tertiary treatment** (chemical purification).

i. Screening and sedimentation: large insoluble particles are removed by passing water through metal screens. The water is then held in large tanks so that suspended solids settle at the bottom under gravity.

ii. Coagulation and flocculation: used, if required, to speed up the settling of finer colloidal particles and to precipitate dissolved contaminants (heavy metals, phosphates). A chemical coagulant such as aluminium sulphate, $Al_2(SO_4)_3$, makes the small particles clump together into larger masses called “floc”, which are easily removed.

iii. Filtration: a mechanical process that removes the remaining fine suspended matter and some microorganisms. Water passes through layers of graded sand and gravel; the sand traps particles in the pores between grains. Activated carbon is used for more advanced purification.

iv. Disinfection: filtration cannot remove dissolved ions or microscopic pathogens, so chemical purification is needed. Chlorination is the most common method; sodium hypochlorite (NaOCl), ozone (O_3) or hydrogen peroxide (H_2O_2) are also used. Chlorine or sodium hypochlorite forms hypochlorous acid (HOCl), which penetrates the cell walls of bacteria and viruses and disrupts their functions.

$$Cl_{2(g)} + H_2O_{(l)} \rightleftharpoons HOCl_{(aq)} + HCl_{(aq)}$$

v. Reverse osmosis (RO) and ion exchange: used for high-purity water (laboratories, regions of high salinity). In RO, water is forced under pressure through a semi-permeable membrane that filters out dissolved salts, bacteria and other impurities. Ion exchange “softens” hard water: the Ca^{2+} and Mg^{2+} ions are swapped for Na^+ or H^+ ions in an ion-exchange column.

The textbook numbers both Disinfection and Reverse Osmosis as “iv”. Figure 33.4 shows the plant sequence: coagulation → flocculation → sedimentation (sludge removed) → filtration → chlorination and disinfection → treated water.

Q7. Explain the difference between DO, BOD, and COD; How they relate to the water pollution?

DO	BOD	COD
Dissolved oxygen: the amount of oxygen available in water for aquatic organisms to breathe.	Biochemical oxygen demand: the oxygen required by microorganisms to break down organic matter in water over a specific period, usually five days at 20 °C.	Chemical oxygen demand: the oxygen required to chemically oxidise organic and inorganic substances in water using oxidising agents ($KMnO_4$, $K_2Cr_2O_7$, etc.).
Key indicator of water quality and ecosystem health.	Key indicator of organic pollution.	Shows the total pollution load.
High DO = clean, unpolluted water; low DO = pollution or heavy decomposition of organic matter.	High BOD = much organic matter, heavy pollution; it depletes dissolved oxygen and harms aquatic life.	High COD = many oxidisable chemical pollutants; poor water quality, contamination by industrial or domestic waste.

Relation to pollution: as pollution increases, DO falls while BOD and COD rise, as the typical values show:

Water type	DO (mg/L)	BOD (mg/L)	COD (mg/L)
Very pure water	7 – 10	< 1	< 2
Drinking water	> 6	< 2	< 5
Moderately polluted	4 – 6	2 – 8	10 – 20
Treated sewage	2 – 4	10 – 30	30 – 100
Raw sewage	< 1	100 – 400	200 – 1000
Industrial waste	0 – 2	500 – 5000+	1000 – 20,000+

QUICK CHECK QUESTIONS (IN-CHAPTER)

These questions appear in the "Quick Check" boxes inside the chapter, not in the end-of-chapter exercise.

Quick Check 33.1

a) Explain how the polarity of water facilitates the transport of agricultural pollutants like pesticides into the water cycle.

Water molecules are permanent dipoles, so water dissolves ionic substances and polar organic compounds by surrounding them. Many pesticides and herbicides therefore dissolve in rain and irrigation water and are carried with it into rivers, lakes and groundwater, where they persist and bioaccumulate.

b) Differentiate between Biological Oxygen Demand (BOD) and Chemical Oxygen Demand (COD) as indicators of water quality.

BOD	COD
Oxygen required by microorganisms to break down organic matter (usually 5 days at 20 °C).	Oxygen required to chemically oxidise organic and inorganic substances using oxidising agents ($KMnO_4$, $K_2Cr_2O_7$).
Indicates the level of organic (biodegradable) pollution.	Indicates the total load of oxidisable pollutants.

Quick Check 33.2

a) Classify these as point or non-point sources of water pollution, explain your choice.
i) The outfall pipes used to release treated sewage into a bay or river from wastewater treatment plants.
ii) The "acid rain" caused by the nitrogen oxides (NO_x) from distant traffic or power stations dissolved in rainwater.

i) Point source: the sewage enters the water from a single, identifiable location (the pipe).

ii) Non-point source: the gases come from many distant, scattered sources and the acid falls with rain over a wide area; no single point of entry can be identified.

b) Give two major differences between point and non-point pollution sources.

Point source	Non-point source
Enters from a single, identifiable location (pipe, tank).	Comes from many diffuse places over a wide area.
Pollutant is concentrated at the site of entry (pollution plume, acute toxicity).	A mixture of pollutants at lower concentrations.
Does not depend on the weather.	Depends on rainfall, snowmelt and storms.
Example: factory wastewater pipe.	Example: farmland and road run-off.

Quick Check 33.3

i. Name the sources of heavy metals.

Industrial discharge, mining and improper waste disposal. Mercury: coal combustion, gold mining, thermometer and battery manufacturing. Lead: old lead pipes, lead-acid batteries, paints (and formerly leaded petrol). Cadmium: metal plating, plastic stabilisers and phosphate fertilizer impurities.

ii. How oil spill is hazardous to the marine life?

Oil from leaks and spills of ships spreads rapidly over the sea surface, forming a layer that blocks sunlight and reduces oxygen exchange. This severely harms fish, birds and mammals: marine mammals ingest the toxic oil while grooming, and fish suffer long-term health problems from the hydrocarbons. Oil also damages coastal habitats such as mangroves, coral reefs and beaches.

iii. Pesticides like DDT are "non-polar and insoluble in water" yet they "pollute water and cause severe human health issues". Explain.

Although DDT does not dissolve in water, it is carried into water bodies by agricultural run-off and it persists there (it is a persistent organic pollutant). Being non-polar, it dissolves in the fatty tissues of fish and birds, accumulates and is magnified up the food chain. Humans who eat these organisms receive high doses, which cause hormonal disruption and are linked to cancers.

Quick Check 33.4

a) A river passes by three regions in a country: a rural area with vast agricultural land and a lot of cattle; an industrial area with many paint and battery industries; and a large residential area of a big city:
i) Predict which type of pollutants will be added to the river in the rural area; organic, inorganic, or both? Name two of them.
ii) Which metals are most likely to pollute the water near the industrial area?
iii) Name the regions that will contaminate the water with chemicals that lead to Eutrophication.
iv) Name four contaminants that will pollute the water near the residential area.

i) Both. Organic: pesticides and animal (cattle) waste. Inorganic: fertilizers such as ammonium nitrate and phosphates.

ii) Lead (paints, lead-acid batteries), together with mercury and cadmium (battery manufacturing).

iii) The rural area (nitrogen and phosphorus from fertilizers and animal waste) and the residential area (sewage and detergents).

iv) Sewage (with bacteria and viruses), detergents, oils and household chemicals, and plastics/solid waste.

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Quick Check 33.5

a) Methylmercury ($[CH_3Hg]^+$) is "highly fat-soluble," allowing it to cross biological membranes and the blood-brain barrier.
i. Contrast the chemical structure of the methylmercury cation ($[CH_3Hg]^+$) with the inorganic mercury ion (Hg^{2+}). Explain why the presence of the methyl group ($-CH_3$) alters its solubility in lipid bilayers of living cells compared to the inorganic ion.
ii. Predict the impact on the "biomagnification" process if ($[CH_3Hg]^+$) was highly water-soluble but had low fat solubility.

i. Hg^{2+} is a simple ion with a 2+ charge; it is strongly hydrated by polar water molecules and cannot dissolve in non-polar lipids. In $[CH_3Hg]^+$ a non-polar methyl group is covalently bonded to mercury and the charge is only 1+. The non-polar $–CH_3$ group gives the ion an organic (lipophilic) character, so it dissolves in the non-polar lipid bilayer and passes through membranes.

ii. It would stay dissolved in body water and be excreted (e.g. in urine) instead of being stored in fatty tissue. It would not accumulate in organisms, so its concentration would not increase up the food chain; biomagnification would be very small.

The explanation uses the "like dissolves like" principle; the chapter states only that methylmercury is highly fat-soluble.

b) How the water pollution can contribute to the global warming?

Sewage, agricultural run-off and industrial waste boost the growth of algae and other organisms. When these die and decompose in oxygen-poor (anaerobic) conditions, methane is released. Methane is a potent greenhouse gas, so it adds to climate change and global warming.

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